

Construction worker detection in video frames for initializing vision trackers

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ABSTRACT

Monitoring the location of resources on large scale, congested, outdoor sites can be performed more efficiently with vision tracking, as this approach does not require any pre-tagging of resources. However, the greatest impediment to the use of vision tracking in this case is the lack of detection methods that are needed to automatically mark the resources of interest and initiate the tracking. This paper presents such a novel method for construction worker detection that localizes construction workers in video frames. The proposed method exploits motion, shape, and color cues to narrow down the detection regions to moving objects, people, and finally construction workers, respectively. The three cues are characterized by using background subtraction, the histogram of oriented gradients (HOG), and the HSV color histogram. The method has been tested on videos taken in various environments. The results demonstrate its suitability for automatic initialization of vision trackers.

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1. Introduction

Tracking data is useful for site monitoring tasks including productivity measurement, activity sequence analysis, and safety management. Radio frequency technologies, such as Radio Frequency Identification (RFID), Global Positioning Systems (GPS) and Ultra Wideband (UWB) have been used to generate real-time data on the location of construction entities across time. The tests and applications of these technologies proved their excellent performance and suitability to construction sites. Constant efforts on the practical use of these technologies for site monitoring resulted in their actual applications on the construction sites such as tracking of heavy equipment [1–3] and precast concrete elements [4]. However, it is necessary to install a sensor on each entity to track when using these technologies. This requirement hinders their application in large scale, congested construction sites where a large number of entities need to be tagged.

As an alternative to these technologies, vision-based tracking has been investigated. It can track multiple entities as long as they are present in camera views. Since the technology works on the basis of video data, it fails to track entities when information in the video data is not sufficient. For example, it does not provide the position of occluded entities (no information) and is not suitable for tracking at night time (limited information). However, it still have the advantage that it can track a large number of entities without the need for tags, which is beneficial for tracking in large scale, congested sites. Hybrid systems combining vision based tracking with other radio

frequency technologies can also be a feasible option in which the strength of each technology can compensate each other's weakness. For example, while RFID or UWB is obviously the best to use for tracking in indoor environments, vision based tracking system can be a potential option for outdoor environments especially in early periods of a project when the site is usually a widely open area with a large number of entities moving around. Also, vision based tracking can provide the data that RFID or GPS has missed because of the limited number of tags or sensors.

Past research efforts showed the potential of vision based systems in tracking construction entities such as workers and concrete buckets. However, the results were limited to 2D pixel coordinates which are generally not sufficient for monitoring construction sites. Recently, a 3D vision-based tracking method was proposed, which provides 3D coordinates of construction resources by using two or more fixed cameras [5]. The framework involves several processes: detecting construction entities, 2D tracking, and calculating 3D coordinates. It detects construction entities in each view, and tracks the detected entities using 2D vision trackers which provide 2D pixel coordinates across frames. The 2D pixel coordinates from multiple camera views are correlated to calculate its 3D coordinates. The authors have proposed and validated methods for 2D tracking and the calculation of 3D coordinates in the past. However, automatic detection of construction entities was not achieved.

In order to initiate a vision tracker, it is necessary to first determine which entities to track. All entities that enter the camera views have to be captured and marked to trigger a vision tracker. Moreover, a vision tracker generally fails to track an entity when it is fully occluded. Therefore, when the entity gets free from the occlusion, it has to be initialized again. Given a pixel region corresponding

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to each entity, a vision tracker extracts the visual patterns from the region and starts tracking on the basis of the patterns. However, it is time consuming and error prone to manually mark all construction entities in multiple views, due to the large amount of entities to track in construction sites. Therefore, an automated way of identifying and localizing the entities is required. Several approaches have recently been employed to detect construction entities in images [6–8]. However, it is not proven whether the methods can differentiate between construction entities and non-construction entities.

This paper presents a novel method that detects and marks construction workers in video frames for the purpose of initializing a vision tracker. The method utilizes three features: motion, shape, and color. These features are used in a sequence of separate steps, narrowing candidate regions down to the regions of moving objects, people, and workers one after another. First, the method recognizes the foreground blobs that represent moving objects. Since the camera positions are fixed, the static background pixel areas are filtered out using background subtraction leaving behind only moving object areas (foreground blobs). Second, human body shapes are modeled through the training of HOG (Histogram of Oriented Gradients) features and searched on the foreground blobs. Third, once human body shapes are detected, they are further processed to be classified into construction workers and non-workers. In this classification process, color histograms which are constructed on the HSV (Hue, Saturation, and Value) color space, are used. The proposed method is tested on videos recorded with people wearing various combinations of safety gear. It is worthwhile to clarify that the proposed method aims to detect construction workers only as soon as they newly appear in the camera view. Therefore, performance is measured by precision and the lapse between the first presence of an entity and its detection. The minimum precision rate higher than 99% and the maximum lapse less than 1 s, validate the suitability and effectiveness of the method for initializing the tracking of construction workers.

2. Background

2.1. State of practice in monitoring resources

In recent years, state-of-the-art technologies are employed in practice for monitoring construction resources. GPS is the most famous technology which is generally used to track a fleet of heavy equipment. It has become a well-established monitoring system in construction sites. The system consists of a constellation of satellites, GPS sensors mounted on each equipment asset, and a central module that communicates with the sensors. Each sensor captures the asset's location and transmits the data to the central module over the mobile network, which makes it possible to visualize the location of all equipment on a single map in real time [9,10]. GPS has usually been applied to outdoor construction practices like positioning of construction equipment [11] and vehicles [12]. However, the maximum error of general GPS is about 10 m, and it can be reduced to 1 cm by using kinematic GPS with trade off in increased costs [13]. Indoor GPS has also been introduced to tracking in indoor construction environments [14]. The system replaces satellites with four or more indoor transmitters. Numerous products of the GPS system have been emerged for tracking and monitoring construction equipment [1–3]. Equipment manufacturers are adopting the products to provide for their customers [15]. The GPS sensors are capable of reporting locations as well as idle and work time, and odometer readings. Analyzing the reported data can bring about significant benefits for contractors such as reduced fuel budget, better equipment utilization, and timely maintenance, etc. [9]. Despite its high accuracy, significant time and effort to deploy the system and expensive cost limit the application. Also, a recent report states that construction business in the United States will face serious disruption to their GPS-involved operations because of 4G-LTE open wireless broadband network that incorporates nationwide satellite coverage [16].

RFID-based systems are also widely used in construction projects. The systems consist of RFID tags, a reader, and a data management system. When a reader requests, each tag sends its own ID so that the reader can identify the unique object. It is generally used to access and track construction materials or workers. When a critical piece of material arrives on site or it is placed in the structure, a tag attached to or embedded in the piece is read to register the event with its ID and timestamp of accomplishment [17]. It has been applied to quality management [18], prevention of collision accident [19], and automatic identification of construction concepts [20]. Recently, Ko [21] proposed a methodology of 3D sensing with RFID which locates the positions of various construction entities. However, the near-sighted effect prohibits its use in tracking applications. There have also been efforts to integrate RFID and GPS technology. Ergen et al. [22] applied this combination to track precast pieces in a storage yard. These research efforts led to the practical use of RFID technologies in a construction project. Engineering News-Record reported about a notable case in which Skanska made use of RFID technologies to track pre-cast structural elements from casting to assembly in a football stadium construction [4]. In the project, RFID tagging was integrated with BIM (Building Information Modeling). Also, an RFID-based system that combines tags on ID cards with a digital camera in the reader to verify identities was reported to be used in more than 400 projects around the world [17,23].

Remote-controlled web-based cameras are currently available and used for remote monitoring of construction sites [24–26]. The cameras are controlled remotely through pan/tilt/zoom functions, and transmit video frames wirelessly to the central system. The wireless communication system enables one person to monitor an entire site, which substantially reduces security cost [24].

2.2. State of research in vision tracking

Both GPS-based and RFID-based systems are based on radio technologies. These radio technologies require the installation of tags on the entities to track, which makes them disadvantageous to tracking in large scale, congested sites [5]. This leads to the increasing interest for vision technologies in construction. The growing number of cameras available in construction sites adds the value and potential of vision technologies. Aiming to achieve tracking on demand with minimum cost and human interaction, vision-based tracking methods are investigated and introduced to construction. These methods track entities in a sequence of video frames and do not require installing any sensors or tags on the entities [27]. They are capable of tracking a large number of entities with a high level of accuracy [28]. Also, they can help prevent the controversial issues associated with construction workers who may be uncomfortable being tagged with sensors.

Teizer et al. [29] investigated vision trackers for construction worker tracking, and Yang et al. [30] proposed a vision tracker that can track multiple construction workers. Gong and Caldas [31] validated that vision tracking can be applied to automate productivity analysis. However, in these works, the tracking data was limited to 2D pixel coordinates and the entities to be tracked were manually selected and marked. Extending the general 2D vision trackers, a framework of 3D vision tracking has recently been presented [5]. It provides 3D coordinates using multiple onsite cameras. It consists of three main steps: 1) automatic initialization of object regions, 2) 2D tracking in each camera, and 3) correlating two camera views to calculate 3D coordinates. Regarding step 2), Park et al. [27] performed a comparative study to find the best tracker for construction. Also, the promising performance of step 3) is fully evaluated and validated [28]. For the first step, the detailed investigation on the automated detection of construction entities is needed.

2.3. Vision-based object detection

The purpose of object detection is to recognize and localize an object category (e.g. face, vehicle, and animal) by image features which

are common to all objects of the type. The challenge of object detection is to construct a feature template compatible to various appearances of the object category. Shape, color, and motion features are generally used for object detection.

Haar-like features [32,33] and HOG [34,35] are well-known shape features. Both features are on the basis of gradient values, but utilize them in different ways. Haar-like features are vectors of image gradients which are differences of intensities between adjacent regions. In order to detect various appearances of an object type, the features are trained through a machine learning process: Haar-like features of various appearances are extracted from abundant training images and the features are trained with machine learning algorithms. For example, Viola and Jones [32] used vertical and horizontal Haar-like features with an AdaBoost (Adaptive Boosting) algorithm [36] for human face detection. Lienhart and Maydt introduced additional 45° Haar-like features to account for diagonal edges and enhance the detection [33]. Even though training with AdaBoost takes several days depending on the processor specifications, the method allows for real time detection once the training is completed. The HOG feature is a collection of local histograms of gradient directions. It divides an image based on a grid of uniformly spaced blocks. For each block, a histogram is calculated by counting the occurrence of gradient orientations. Similar to Haar-like features, HOG features also need training with a large number of images. Dalal and Triggs [34] applied HOG features trained with SVM (Support Vector Machine) [37] for human detection. They showed the superiority of HOG features over Haar-like features in human detection. Zhu et al. [35] sped up the human detection by using AdaBoost while retaining the equivalent accuracy.

Color is also a useful and intuitive feature for recognizing an object type. A color histogram [38] is one of the typical color features. Simple calculation and invariance to rotation and translation are its useful qualities for object detection. It is suitable for detecting an object in distinctive colors. However, the sensitivity to illumination and the lack of spatial information limits its applications. The color histogram has been broadly used for image segmentation and content-based image retrieval in which spatial information is not critical [39,40]. It has also been employed in tracking applications. In tracking processes, color histograms combined with HOG features were used as observation models of pedestrians [41], hockey players [42], etc. However, color histograms are not appropriate for the detection of pedestrians or hockey players since their colors widely vary from person to person according to their clothing. In Lu et al.'s work [42], Haar-like features are used for detection, and the color histogram is used exclusively for tracking the detected entities.

Background subtraction [43–45] detects moving objects on the basis of their motion cues. This method can be applied only to fixed camera views. The static background scene is modeled by taking a dominant value for each pixel across a certain number of frames. The model has to be updated throughout the frames in order to reflect changes of illumination. McFarlane and Schofield [43] presented an approximated median filter which estimates the background with a simple update process. It increases/decreases the background estimate if the pixel value of the new video frame is larger/smaller than the estimate. The MoG (Mixture of Gaussians) method [44] is one of the most popular background modeling methods. It models the background pixel values as a mixture of multiple Gaussian distributions. Moving objects are detected by thresholding the difference of pixel values between the current frame and the background. Li et al. [45] screened out moving background objects such as wavering trees and shadows by modeling them with the color co-occurrence feature. Background subtraction is computationally efficient since it can detect all moving objects simultaneously regardless of their appearances or types. However, motion features are not suitable to identify object types. Furthermore, background subtraction may recognize multiple objects that partially overlap each other as a single object.

Recently, several approaches for detecting construction entities have been proposed. Park et al. [6] and Jog et al. [7] presented methods

for detection of wheel loaders and trucks, respectively. They used the Semantic Texton Forests approach [48], which detects objects through segmentation and classification. Chi and Caldas [8] proposed a method that detects and classifies construction entities including construction equipment and workers. Several characteristics of foreground blobs which result from background subtraction are trained for the classification.

3. Problem statement and objectives

As described earlier, vision-based tracking has great potential in monitoring large scale, congested sites. However, detection of construction entities is required to automate its initialization process that locates and marks the entities to track. Even though Chi and Caldas [8] proposed a method of detecting construction entities including workers, their work lacks information regarding how to differentiate a worker from a pedestrian. This distinction is vital in construction sites located in residential areas. Moreover, their method is sensitive to illumination changes since it relies only on the foreground blob features. Therefore, the overall objective of this research is to automate the detection of construction workers. Specifically this research aims to 1) investigate image features that effectively characterize the appearance of 'construction worker', 2) determine appropriate measures to evaluate the method regarding that this research is aiming strictly for accurate initialization of vision trackers, and 3) test the method for detecting multiple workers in various illumination conditions.

4. Methodology

In response to the objective stated above, this paper presents a method and its validation for recognizing and locating construction workers in video frames. In this paper, 'construction worker' is considered as a person wearing safety gear such as safety vests and hard hats. The proposed method takes advantage of three types of image features to describe construction workers: motion, shape and color. These features are separately implemented in sequential steps. First, it detects foreground blobs where objects in motion are expected to exist. Given the fixed camera views, the difference of pixel values between a background model and the incoming frames is the main cue to recognize motions. Second, it identifies the regions corresponding to human bodies from the foreground blobs based on their patterns of HOG features. Third, the detection regions that result from the second step are classified into construction workers and non-workers. It uses color histograms and a k-NN (k-Nearest Neighbors) [50] classifier to characterize fluorescent colors of safety gear. Fig. 1 illustrates the described framework. In short, the first and second steps detect people including workers based on their motion and shape features, then the third step sort out workers from the detected people by analyzing their color features.

Background subtraction, in the proposed method, reduces the candidate regions for worker detection. It is used to achieve higher computational efficiency and lower false alarm rates. The second step, which is the detection of people with HOG, is computationally far more expensive than the background subtraction. Background subtraction restricts the search areas of the second step. It allows the step to scan only foreground blobs and prevents from unnecessarily scanning irrelevant areas. In addition, it prevents the detection of non-people objects in the background scene (i.e. a static area which has a similar shape to people) in the following step. It eventually leads to higher precision. Once foreground blobs are obtained, they are post-processed to remove small blobs and to merge adjacent blobs. For each foreground blob, the smallest rectangle that encloses the blob is found. Finally, each rectangle is inflated to allow for some margin around a moving object, which is necessary to calculate HOG shape features in the next step. Various methods of background

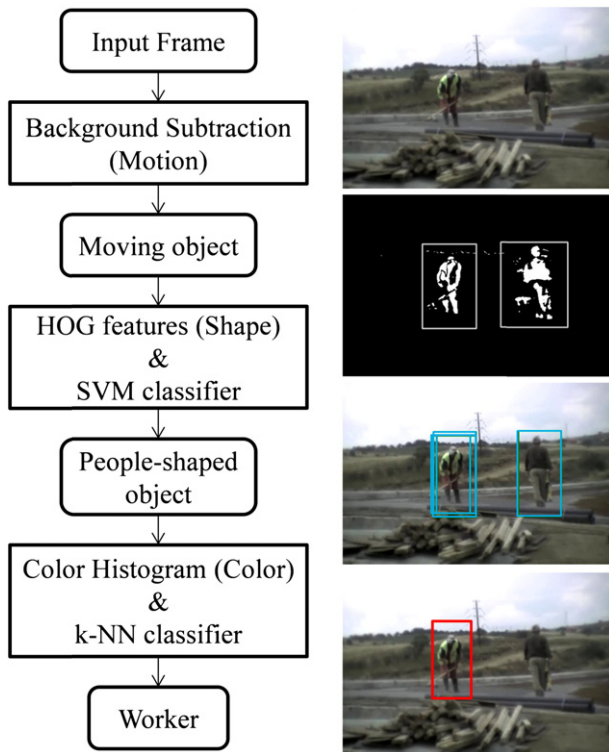


Fig. 1. The framework of the proposed method.

modeling have been presented: approximate median filter [43], MoG [44], color co-occurrence [45], etc. In this research, the median filter method is employed since it is computationally the most efficient. Fig. 2 exhibits the foreground blobs extracted with the three methods and the rectangular foreground regions after post processing. Even with the simplest process, the median filter method provides comparable results to the other methods.

Wearing hard hats or safety vests does not induce significant changes in human shapes. Fig. 3 shows two gradient images obtained by averaging a number of gradient images of ordinary people and construction workers. Human can be intuitively inferred from both gradient images. Therefore, as a step to narrow down candidate regions to people, the second step employs HOG features which have proven to be robust in detecting human bodies. Detection of people in test images works by sliding a search window across the foreground blob regions. The HOG feature is calculated from each window and classified into either people or non-people. In order to embed various appearances of people into the HOG feature template, training of a classifier is necessary. The SVM is chosen for the training as in Dalal and Triggs' work [34]. On the basis of HOG features extracted from a large number of positive (images of people) and negative (images without any people) images, an SVM classifier learns to recognize the patterns of HOG features and to distinguish people from others. Training images have to be in the same size so that all their HOG features have the same vector size.

The results of the second step are the rectangular regions determined to include a person inside. These rectangles are further processed by another classifier which determines whether they are workers or not. In other words, this step filters out the non-workers out of the people regions. The HSV color histogram is selected for this purpose. ANSI (American National Standards Institute) stipulates that construction workers must use high-visibility safety apparel since they are regularly exposed to the hazards of low visibility. It specifies three colors for the high-visibility apparel: fluorescent yellow-green, fluorescent orange-red, and fluorescent red [51]. Since the colors are limited and all fluorescent colors, it is viable to characterize the patterns of the safety gear colors.

The RGB color space, which is an additive color model, is not effective for modeling the safety gear colors because of its sensitivity to illumination conditions. This fact is illustrated in Figs. 4 and 5 which show the RGB color histograms (32 bins) of yellow-green and orange-red safety vests, respectively. In each figure, the two safety vests are the same type, but with different illumination conditions—one in a bright condition (1st row), and the other in a dark condition (2nd row). It is observed that significant disparity between the color histograms in the 1st and 2nd rows is caused by the slight change in illumination. Compared to the RGB color space, the HSV color space is in better accordance with the conceptualization of human eyes, thus, is more appropriate in categorizing objects by colors. The three components of hue, saturation, and value measure the actual color, purity, and intensity (brightness), respectively. Figs. 6 and 7 show the HSV color histograms (32 bins) of the safety vests as in Figs. 4 and 5. Hue and saturation histograms in the 1st and 2nd rows exhibit similar patterns in spite of the illumination change which is reflected in the value histogram. In addition, hue and saturation histograms in Figs. 6 and 7 are distinct from those of ordinary vests on pedestrians in Fig. 8. Fig. 9 shows saturation images of a pedestrian and construction workers. Unlike in the pedestrian image, the safety vest regions in construction worker images are highlighted by the saturation components regardless of their colors. These illustrations signify that the distinct characteristics of safety vests can be effectively modeled with hue and saturation histograms. Accordingly, the third step exploits hue and saturation histograms for characterizing the patterns of the safety gear.

As in the second step, this step also needs to go through a training process. Positive and negative data, in this step, are the images of construction workers and ordinary people, respectively. Focusing exclusively on safety vests and hard hats, this step takes into account only the upper half of the human body to construct color histograms. The obtained color histogram is normalized to minimize the effect of the image window sizes. Negative images contain a few images of people wearing jackets of which the colors are similar to safety vests. These images prevent SVM from constructing clear boundaries between the color histogram vectors of worker and non-worker. The k-NN is more suitable to this case because it performs classification simply based on a majority vote of its k neighbor. Even though the main role of this third step is classifying a person into a worker or a non-worker, it also acts as a filter that eliminates false positive results of the second step (non-people regions).

5. Implementation, experiments and results

The whole framework as a single package is implemented using Microsoft Visual C# in NET Framework 4.0 environment. The size of the HOG template used for people detection is 64×128 . In order to detect people smaller than the template size, the foreground regions are scaled up. The HOG feature is calculated on the basis of two spatial layers (cells and blocks). The first layer is comprised of a grid of cells (8×8 pixels) where a local HOG is obtained. The second layer is comprised of a grid of blocks. A block is a collection of 2×2 cells. Four local HOG's of 2×2 cells in a block are aggregated and normalized together. The normalization allows for invariance to illumination changes. The histogram is constructed based on the angle and magnitude of gradients. It consists of 9 bins which represent evenly distributed angles ranging from 0 to 180° . A gradient magnitude at each pixel is added to the corresponding bin which the angle falls into. A sequence of HOG features that are spatially distributed in a grid manner structures a final descriptor of the input image.

The k-NN classifiers associated with color histograms are trained with images of upper half bodies wearing both hard hats and safety vests. A majority vote of ' k ' nearest neighbors determines the classification. Parameter studies are performed to determine the optimal number of histogram bins as well as the optimal number of nearest

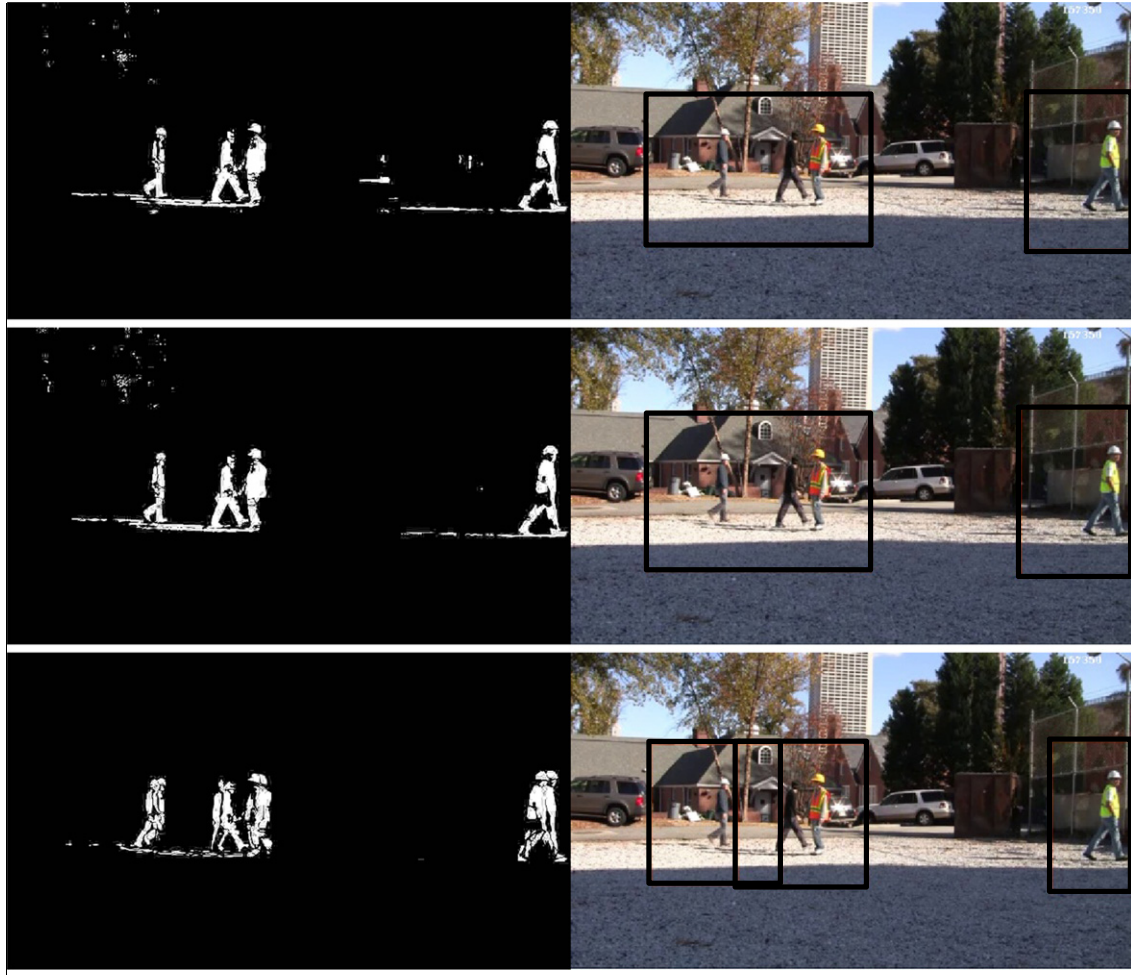


Fig. 2. The foreground regions which result from the approximate median filter (1st row), the MoG (2nd row), and the color co-occurrence (3rd row) methods.

neighbors (k). The results of the studies are presented in Section 5.3. In all performed tests, 500 positive images (upper half body with safety gear), including yellow-green, orange-red, and red safety vests, are collected and used for the training of color histograms.

To validate the implemented method, the method is tested on ten high-definition (HD) videos taken with a HD camcorder (Canon VIXIA HF S100). However, the videos are resized to 768×432 for the sake of fast processes with similar level of performance. 5 frames per second are processed in near real time. The videos are taken of 4–5 people wearing different combinations of safety gear (hard hats and safety vests) to investigate the effect of the combinations on

the detection results. All the videos have different backgrounds and various illumination conditions.

5.1. Metrics for performance evaluation

Two metrics are chosen to measure the performance of the method. The metrics are selected from the perspective that the main purpose of the method is initializing a vision tracker. The first metric is precision which is popular in evaluating pattern recognition. To clarify the meaning of precision, let us define the number of correctly detected workers as TP (true positive), the number of the detections

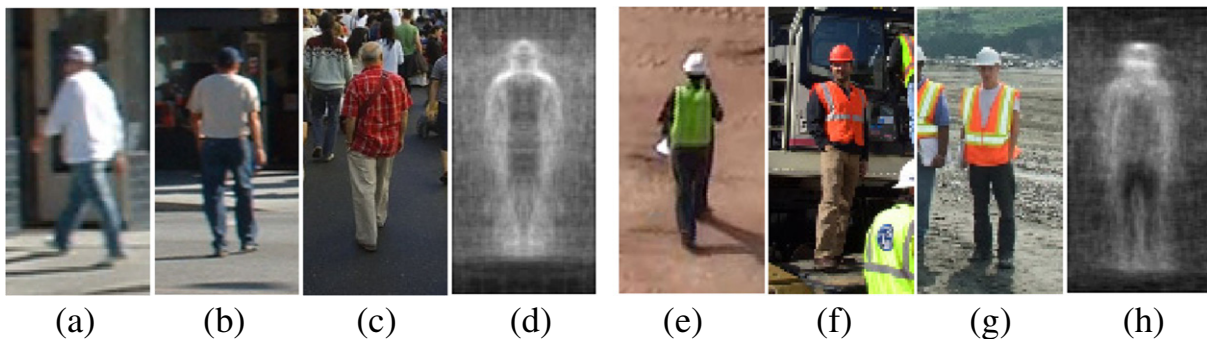


Fig. 3. (a)–(c): examples of people images, (d): average gradients of 200 people images, (e)–(g): examples of worker images, (h): average gradients of 200 worker images.

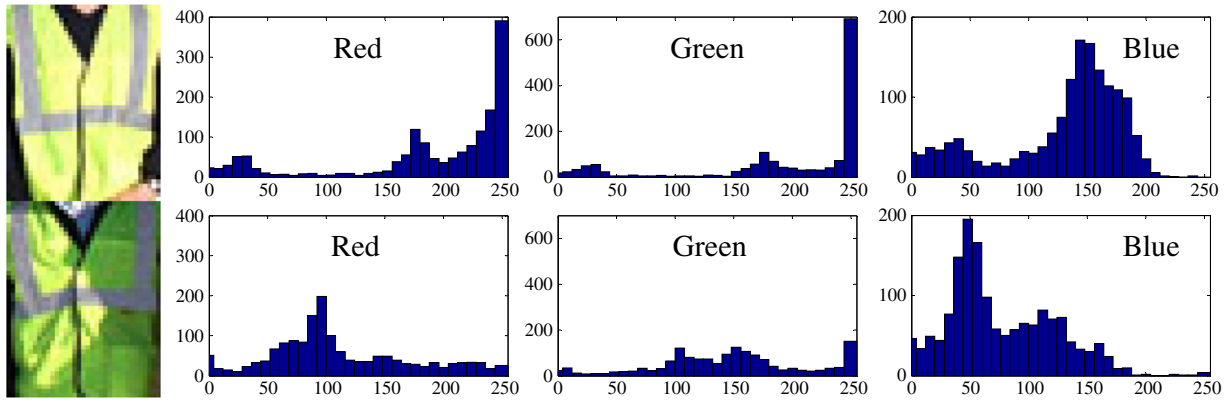


Fig. 4. RGB color histograms of yellow-green safety vests in bright (1st row) and dark (2nd row) illumination conditions.

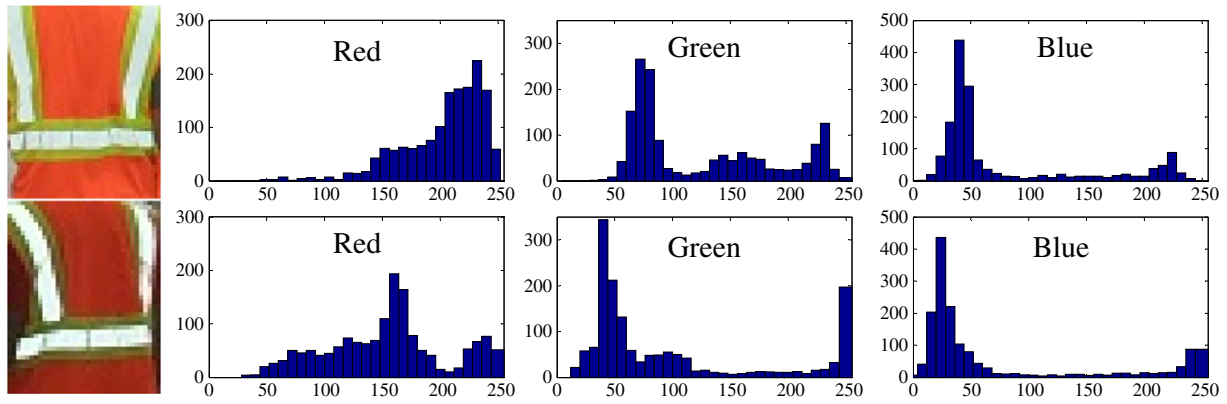


Fig. 5. RGB color histograms of orange-red safety vests in bright (1st row) and dark (2nd row) illumination conditions.

which is not relevant to a worker as FP (false positive), and the number of missed workers (i.e. not detected) as FN (false negative). Precision is defined as a ratio of TP to $TP + FP$ and measures the reliability of the detection. The second metric is the time delay of detection. It measures how instantly it detects a worker after one enters the scene or gets out of an occlusion case. It is worthwhile to note that the recall is not an appropriate measure of a method for vision tracker initialization. The recall is the ratio of TP to $TP + FN$. The proposed method is not designed to detect a worker in every frame, for which a tracker is responsible. For instance, a method that detects a worker

on every other frame resulting in just 50% recall (low recall) could perform well for initializing a tracker as long as it detects no other type than a worker (high precision).

5.2. The definition of 'construction worker'

Although recall is not an appropriate metric overall, it is calculated for the analysis of the effect of safety gear combinations. Table 1 illustrates the combinations involved in three of the videos. Since the method is devised to detect workers that appear in the view,

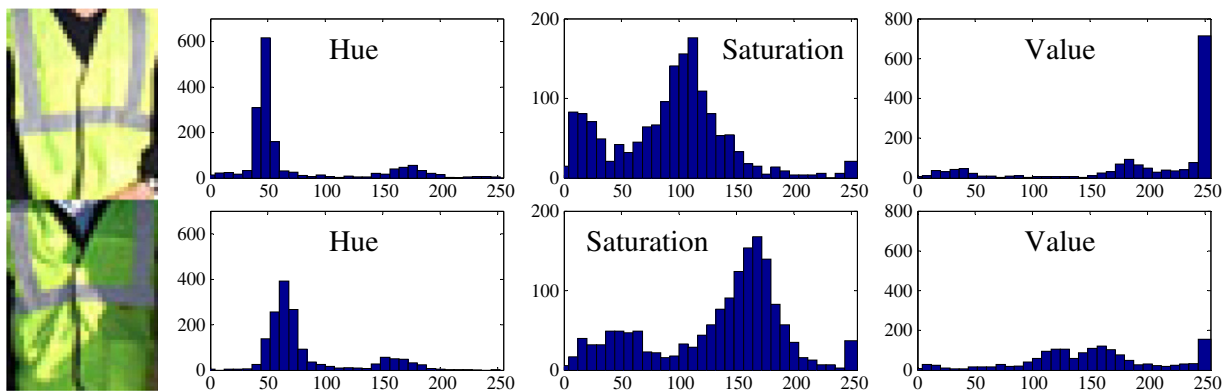


Fig. 6. HSV color histograms of yellow-green safety vests in bright (1st row) and dark (2nd row) illumination conditions.

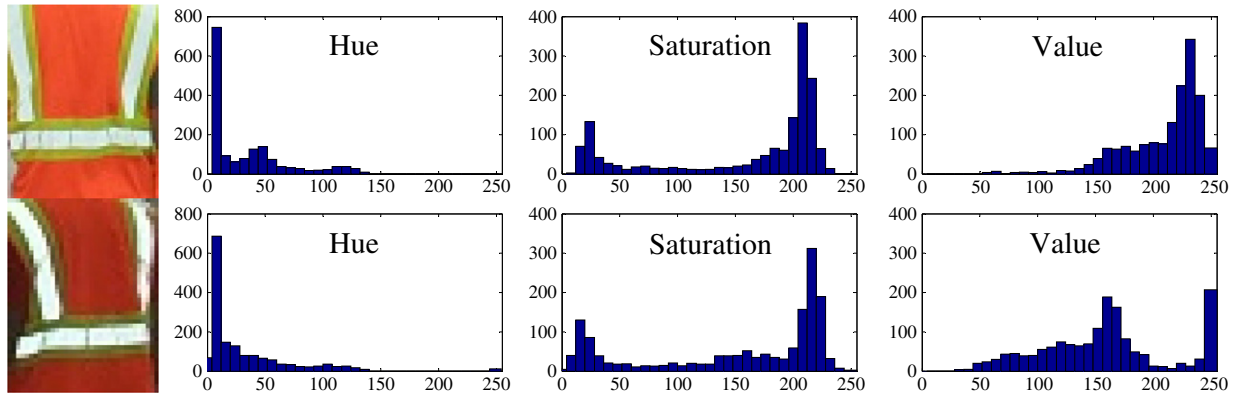


Fig. 7. HSV color histograms of orange-red safety vests in bright (1st row) and dark (2nd row) illumination conditions.

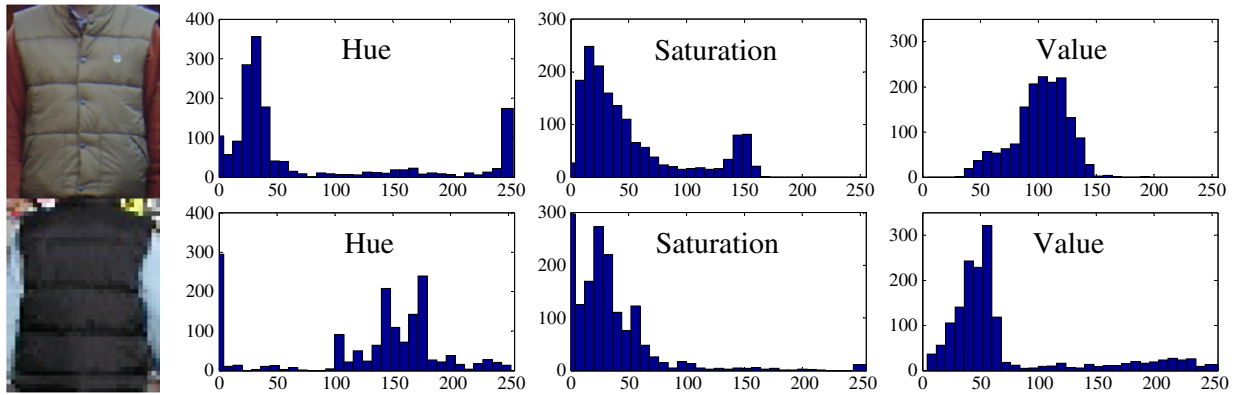


Fig. 8. HSV color histograms of ordinary vests on pedestrians.

occluded workers are not the target of the method. Hence, an occluded person is not counted in this experiment. Table 2 summarizes the recall of five people in three videos. There is clear distinction between Persons 1–2 and Persons 3–5. Persons 1 and 2 who did not wear safety vests are barely detected. Person 3 who only wore a safety vest is detected far more frequently than Persons 1–2, and its detection rate is comparable to Persons 4–5's. It can be inferred from this fact that the safety vest has a more decisive effect on color histograms than the hard hat. As ANSI/ISEA [51] stipulates, safety vests have a more limited number of colors, which contributes to the effective detection. According to these observations, a 'construction worker' is defined in this paper as a person wearing at least a safety vest. Hence, Persons 3, 4 and 5 are defined as construction worker in the following experiments. Since the proposed method first detects people and then sort out construction workers, it can eventually detect

both workers and non-workers and classify them. Fig. 10 exhibits two result frames from each video. Whereas the left images contain only correct detections, the right ones contain false results as well. Persons 4 and 3 are classified as non-worker in Videos 1 and 3, respectively (FN). Person 2 is classified as construction worker in Videos 2 (FP). The third step which plays a core role in classifying construction worker and non-worker takes only 14 ms in average for each frame which is about 9% of the total processing time. This classification will be useful to warn people who are not wearing safety vests on a construction site for safety issues.

5.3. Results—precision and time delay

Precision is calculated according to the definition of 'construction worker', and time delay is also measured for all ten videos. It should



Fig. 9. Saturation images of a pedestrian (a) and construction workers (b and c).

Table 1
The combinations of safety gears.

Person	Video 1		Videos 2, 3	
	Safety vest	Hard hat	Safety vest	Hard hat
1	X	X	X	White
2	X	White	X	White
3	N/A		Orange-red	X
4	Yellow-green	White	Yellow-green	White
5	Orange-red	Yellow	Orange-red	Yellow

X: not wearing.

Table 2
The detection rate (recall) of 5 people.

Recall (%)	Video 1	Video 2	Video 3
Person 1	0.0	0.0	0.0
Person 2	0.0	1.8	6.3
Person 3	–	77.6	71.1
Person 4	88.7	96.9	80.9
Person 5	88.3	74.6	76.8

be noted that several videos include both shade and light where abrupt changes in illumination occur as workers cross the border between them (e.g. Videos 2 and 3 in Fig. 10). Figs. 11–13 show the results of parameter studies on the number of bins in saturation and hue histograms, and the number of nearest neighbor (k). The number

of bins in saturation histograms is determined to be ‘8’ because of its highest precision and the delay less than 1 s (Fig. 11). ‘64’ is selected for hue histograms since it exhibits the minimal delay and comparable precision with other sizes (Fig. 12). Accordingly, the total size of the color histogram is 72 ($=64+8$). Regarding the value of ‘k’ in the k-NN learning process, ‘11’ is found the best (the least delay in Fig. 13).

Using the selected parameters, four tests are conducted for each video through the training of color histograms with 550, 1100, 1650, and 2200 negative images. The results are summarized in Tables 3 and 4. Results of step 1 (background subtraction) and 2 (people detection based on HOG) remain the same in all four tests. In Table 3, precision increases as more negative images are used. It indicates that the use of more negative images helps reduce FP’s. However, on the contrary, recall works inversely to the number of negative images, which means TP’s are also reduced by using more negative images. Fig. 14 shows examples of these effects. The first and second rows illustrate the removal of FP’s (non-person and Person 2) while the third row shows a missed TP (Person 5). Based on precision results which is selected as a critical metric (Section 5.1), the proposed method achieved the best performance when 2200 negative images are used in training the color histograms of upper half body. As explained in the previous section, time delays are considered to better reflect the performance of the initialization method. In total, 73 cases are observed in which a worker enters a view or gets free from occlusion. In Table 4, the use of 2200 negative images results in a maximum of 0.67 s delay which is only 0.11 s longer than using 550 negative images. The maximum is



Fig. 10. Detection of people who are wearing safety vests (red) and not wearing safety vests (green) (Videos 1–3 from top to bottom).

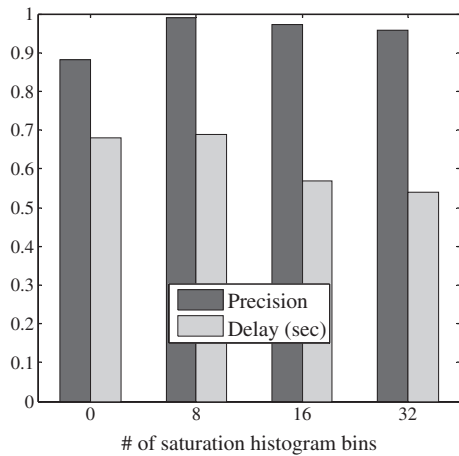


Fig. 11. The performance variation of the method depending on the number of bins in a saturation histogram.

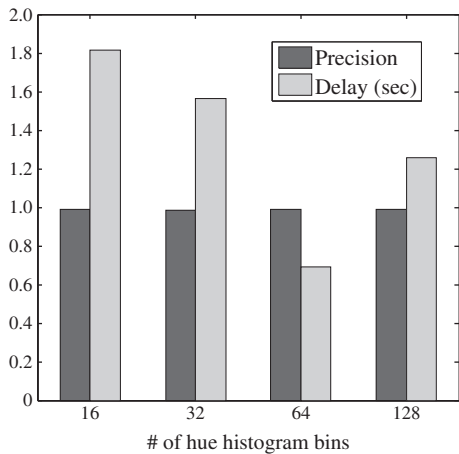


Fig. 12. The performance variation of the method depending on the number of bins in a hue histogram.

calculated with a 95% confidence interval for the standard normal distribution. The 99.0% precision and the 0.67 s delay signify the potential of the proposed method for initializing a vision tracker.

Additionally, since the proposed method is composed of three sequential steps, the performance of each step is further investigated. For this purpose, the critical cause of the missed workers (FN) is

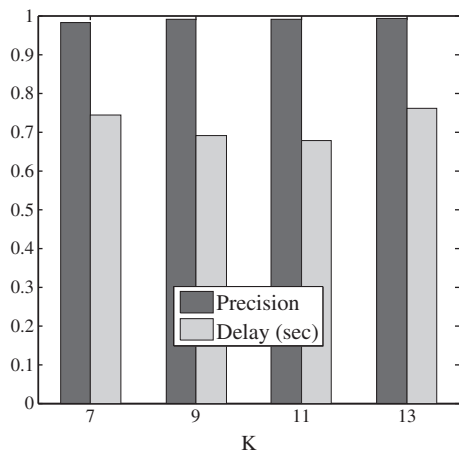


Fig. 13. The performance variation of the method depending on the 'K' value.

Table 3

The precision of detection results.

# of negative training images	550	1100	1650	2200
Precision (%)	90.1	97.2	97.7	99.0
Recall (%)	87.1	83.6	82.4	81.4

analyzed (Table 5). The FN cases are classified into the three steps which cause them. Overall, the third step is discovered as a main step that filters out FN's. In detail, it is also observed that Persons 3 and 5, who wear orange-red vests, are mainly discarded by the third step while Person 4, who wears a yellow-green vest and a white hard hat, is discarded in the second step. The former is attributed to the negative images of people with orange-red shirts. In fact, negative images of upper half body contain a considerable number of images with orange or red shirts which could be confused with orange-red safety vests. In addition, white is a dominant color of hard hats in the positive images which force to eliminate Person 3 (brown hair) and Person 5 (yellow hard hat). On the contrary, it is inferred that the latter results from the similar background color to Person 4. Similar colors of background make it difficult to extract a human body shape.

6. Conclusions

Onsite tracking provides the trajectories of construction entities which are useful for productivity measurements, activity sequence analysis, and site safety management. Popular tracking solutions such as GPS and RFID are mainly based on Radio Frequency technologies. They have an established record of tracking resources for various construction scenarios. However, the requirement for a separate sensor on each entity to be tracked limits their applicability in large scale, congested construction sites. For such cases, vision tracking has recently been proposed as an alternative solution. It has the potential to track a large number of construction entities only with a small number of cameras. The major hindrance to the use of vision tracking is the lack of detection methods that are needed to automatically initiate the tracking. Even though several methods of detecting construction entities were proposed, no approach has been made to differentiate construction workers from pedestrians. Also, a robust method against illumination changes is missing. This paper presents a novel method for automating the detection of construction workers in the video frames. The method exploit background subtraction, HOG shaper features, and color histograms one after another. Background subtraction recognizes the foreground blobs of moving objects, and HOG features are used to detect people from the recognized blobs. The detected people are classified into workers and non-workers on the basis of their color histograms. Two metrics are selected to evaluate its performance. The first is precision, and the second is the time delay of initialization. The proposed method aims to maximize precision

Table 4

The delay of detection.

# of negative training images		Frames	Seconds
550	Average	1.5	0.30
	Standard Deviation	0.8	0.16
	Maximum	2.8	0.56
1100	Average	1.6	0.31
	Standard Deviation	1.0	0.20
	Maximum	3.2	0.64
1650	Average	1.6	0.33
	Standard Deviation	1.1	0.21
	Maximum	3.4	0.68
2200	Average	1.6	0.32
	Standard Deviation	1.1	0.21
	Maximum	3.4	0.67



Fig. 14. Detection of construction workers using 1100 (left) and 2200 (right) negative images.

Table 5

The reason for the missed detection.

Step	Background Subtraction	HOG shape detection	Color histogram classification
Person 3	9.2%	26.5%	64.3%
Person 4	0.9%	72.7%	26.4%
Person 5	17.2%	13.2%	69.5%
Total	10.5%	33.8%	55.8%

and minimize the time delay. The proposed method is preliminarily tested for detecting five people wearing different combinations of safety gear. Based on the preliminary tests, 'construction worker' is defined as a person wearing a safety vest. Experiments are conducted on ten videos with different backgrounds. According to the definition of 'construction worker', the experiments resulted in 99.0% precision and 0.67 s time lapse. These results signify that the proposed method can effectively initialize the tracking of construction workers.

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References

- [1] L. Cable, Construction equipment monitoring system, Government Product News, Penton Media, December 2010 (<http://govpro.com/products/fleets/management/construction-equipment-monitoring-201012/>).
- [2] Engineering News-Record, Equipment Monitor: Wireless Locator, The MacGraw-Hill Companies, July 2008 (http://enr.construction.com/products/newProducts/archives/product_snapshot/2008/1203-productsnapshot.asp).
- [3] Engineering News-Record, Equipment Tracking: Wireless Monitoring, The MacGraw-Hill Companies, December 2008 (http://enr.construction.com/products/product_snapshot/2008/1203-productsnapshot.asp).
- [4] T. Sawyer, \$1-billion Jigsaw Puzzle Has Builder Modeling Supply Chains, Engineering News-Record, The MacGraw-Hill Companies, April 2008 (<http://enr.construction.com/features/technologyEconst/archives/080423-1.asp>).
- [5] I. Brilakis, M.W. Park, G. Jog, Automated vision tracking of project related entities, Journal of Advanced Engineering Informatics 25 (4) (2011) 713–724.
- [6] M.W. Park, G. Jog, I. Brilakis, Initializing vision based trackers using semantic texton forests, The 28th International Symposium on Automation and Robotics in Construction, Seoul, Korea, Proceedings, 2011.
- [7] G. Jog, M.W. Park, I. Brilakis, Truck face recognition using semantic texton forests, The ASCE Construction Research Congress, Ontario, Canada, Proceedings, 2011.
- [8] S. Chi, C.H. Caldas, Automated object identification using optical video cameras on construction sites, Computer-Aided Civil and Infrastructure Engineering 26 (5) (2011) 368–380.
- [9] R. Eecke, Greening (and saving) with GPS fleet tracking, Construction News Tracker, Cygnus Business Media, 2010 (<http://www.forconstructionpros.com/article/10288916/greening-and-saving-with-gps-fleet-tracking>).
- [10] H. Henderson, Benefits of machine control, Construction Contractor, December 2008.
- [11] A.A. Oloufa, M. Ikeda, H. Oda, Situational awareness of construction equipment using GPS, wireless and web technologies, Automation in Construction 12 (6) (2003) 737–748.
- [12] M. Lu, W. Chen, X.S. Shen, H.C. Lam, J.Y. Liu, Positioning and tracking construction vehicles in highly dense urban areas and building construction sites, Automation in Construction 16 (5) (2007) 647–656.
- [13] C.H. Caldas, D.G. Torrent, C.T. Haas, Integration of automated data collection technologies for real-time field materials management, The 21st International Symposium on Automation and Robotics in Construction, Jeju, Korea, Proceedings, 2004.

- [14] H.M. Khoury, V.R. Kamat, Evaluation of position tracking technologies for user localization in indoor construction environments, *Automation in Construction* 18 (4) (2009) 444–457.
- [15] Construction Equipment, DPL America, Hyundai selects DPL's asset monitoring solution, *Construction Equipment*, December 2010 (<http://www.constructionequipment.com/hyundai-selects-dpl%E2%80%99s-asset-monitoring-solution>).
- [16] ForConstructionPros.Com, Study shows interference with GPS poses \$96 billion threat to US economy, *Cygnus Business Media*, June 2011 (http://www.forconstructionpros.com/press_release/10366178/study-shows-interference-with-gps-poses-96-billion-threat-to-us-economy).
- [17] Engineering News-Record, Behind a Murky Name Lurks an Amazing Little Tag, The MacGraw-Hill Companies, April 2008 (<http://enr.construction.com/opinions/editorials/archives/080423.asp>).
- [18] L.C. Wang, Enhancing construction quality inspection and management using RFID technology, *Automation in Construction* 17 (4) (2008) 467–479.
- [19] T. Elghamrawy, F. Boukamp, Managing construction information using RFID-based semantic contexts, *Automation in Construction* 19 (8) (2010) 1056–1066.
- [20] S. Chae, T. Yoshida, Application of RFID technology to prevention of collision accident with heavy equipment, *Automation in Construction* 19 (3) (2010) 368–374.
- [21] C.H. Ko, RFID 3D location sensing algorithms, *Automation in Construction* 19 (5) (2010) 588–595.
- [22] E. Ergen, B. Akinci, R. Sacks, Tracking and locating components in a precast storage yard utilizing radio frequency identification technology and GPS, *Automation in Construction* 16 (3) (2007) 354–367.
- [23] T. Sawyer, South Korean research in electronic tagging is forging ahead, *Engineering News-Record*, The MacGraw-Hill Companies, April 2008 (<http://enr.construction.com/features/technologyEconst/archives/080423a.asp>).
- [24] K. Gomez, Mobile security + remote monitoring, *Construction Contractor*, November 2008.
- [25] Engineering News-Record, Solor-powered Webcam: Remotely Monitor Job Sites, The MacGraw-Hill Companies, 2008 (<http://enr.construction.com/products/newProducts/archives/080730.asp>).
- [26] Engineering News-Record, Jobsite Monitoring: Remote Camera Keeps an Eye on Projects, The MacGraw-Hill Companies, 2008 (<http://enr.construction.com/equipment/features/archives/080301-26-2.asp>).
- [27] M.W. Park, A. Makhmalbaf, I. Brilakis, Comparative study of vision tracking methods for tracking of construction site resources, *Automation in Construction* 20 (7) (2011) 905–915.
- [28] M.W. Park, C. Koch, I. Brilakis, 3D tracking of construction resources using an on-site camera system, *Journal of Computing in Civil Engineering-ASCE* 26 (4) (2012) 541–549.
- [29] J. Teizer, P.A. Vela, Personnel tracking on construction sites using video cameras, *Advanced Engineering Informatics* 23 (4) (2009) 452–462.
- [30] J. Yang, O. Arif, P.A. Vela, J. Teizer, Z.K. Shi, Tracking multiple workers on construction sites using video cameras, *Advanced Engineering Informatics* 24 (4) (2010) 428–434.
- [31] J. Gong, C.H. Caldas, Computer vision-based video interpretation model for automated productivity analysis of construction operations, *Journal of Computing in Civil Engineering* 24 (3) (2010) 252–263.
- [32] P. Viola, M. Jones, Rapid object detection using a boosted cascade of simple features, 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition, Vol. 1, Proceedings, 2001, pp. 511–518.
- [33] R. Lienhart, J. Maydt, An extended set of Haar-like features for rapid object detection, 2002 International Conference on Image Processing, Vol. 1, Proceedings, 2002, pp. 900–903.
- [34] N. Dalal, B. Triggs, Histograms of oriented gradients for human detection, 2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition, Vol. 1, Proceedings, 2005, pp. 886–893.
- [35] A. Zhu, M.C. Yeh, K.T. Cheng, S. Avidan, Fast human detection using a cascade of histograms of oriented gradients, IEEE Computer Society Conference on Computer Vision and Pattern Recognition, Proceedings, 2006, pp. 1491–1498.
- [36] Y. Freund, R.E. Schapire, A decision-theoretic generalization of on-line learning and an application to boosting, *Journal of Computer and System Sciences* 55 (1) (1997) 119–139.
- [37] T. Joachims, Making large-scale SVM learning practical, *Advances in Kernel Methods—Support Vector Learning*, MIT Press, Cambridge, MA, USA, 1999, pp. 169–184.
- [38] M.J. Swain, D.H. Ballard, Color indexing, *International Journal of Computer Vision* 7 (1) (1991) 11–32.
- [39] Z.H. Zhang, W.H. Li, B. Li, An improving technique of color histogram in segmentation-based image retrieval, Fifth International Conference on Information Assurance and Security, Vol. 2, Proceedings, 2009, pp. 381–384.
- [40] C.B. Huang, S.S. Yu, J.L. Zhou, H.W. Lu, Image retrieval using both color and local spatial feature histograms, 2004 International Conference on Communication, Circuits, and Systems, Vols. 1 and 2, 2004, pp. 927–931.
- [41] H. Sugano, R. Miyamoto, Parallel implementation of pedestrian tracking using multiple cues on GPGPU, 2009 IEEE 12th International Conference on Computer Vision Workshops, Proceedings, 2009, pp. 900–906.
- [42] W.L. Lu, K. Okuma, J.J. Little, Tracking and recognizing actions of multiple hockey players using the boosted particle filter, *Image and Vision Computing* 27 (1–2) (2009) 189–205.
- [43] N.J.B. McFarlane, C.P. Schofield, Segmentation and tracking of piglets in images, *Machine Vision and Applications* 8 (3) (1995) 187–193.
- [44] C. Stauffer, W.E.L. Grimson, Learning patterns of activity using real-time tracking, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 22 (8) (2000) 747–757.
- [45] L.Y. Li, W.M. Huang, I.Y.H. Gu, Q. Tian, Foreground object detection in changing background based on color co-occurrence statistics, Sixth IEEE Workshop on Applications of Computer Vision, Proceedings, 2002, pp. 269–274.
- [48] J. Shotton, M. Johnson, R. Cipolla, Semantic texton forests for image categorization and segmentation, 2008 IEEE Conference on Computer Vision and Pattern Recognition, Proceedings, 2008, pp. 1–8.
- [50] T. Cover, P. Hart, Nearest neighbor pattern classification, *IEEE Transactions on Information Theory* 13 (1) (1967) 21–27.
- [51] ANSI/ISEA, ANSI/ISEA 107–2010: American national standard for high-visibility safety apparel and headwear, International Safety Equipment Association (ISEA), 2010.